TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 6
ICT and Society



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 6 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 7, materials, a quiz, a poster rubric
- **Answers** - chapters 1 to 3, in the same order as the workbook
- **Unit 6 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ A computer with the Internet and a projector for the e-learning lessons.
- ☐ A few safe old items to show e-waste: a keyboard, a mouse, a cable, a CD.
- ☐ Newspapers with job advertisements for Practical 6.

"Show your students how ICT helps their village, their school and their future."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Which field?	8
	B Fill in the blanks	8
	C Write two services	8
	D Choose the correct answer	5
	E True or false	6
Chapter 2	A Name the e-waste	8
	B Match the poison	6
	C Which part of the body?	6
	D Reduce, reuse or recycle?	8
	E True or false	6
	F Fill in the blanks	8
	G Choose the correct answer	5
	H Word search	10
Chapter 3	A Match the jobs	8
	B Who should do it?	6
	C True or false	6
	D Choose the correct answer	5
	E My dream ICT job	4
	F Answer these	4
	G Read the advertisement	5
	H Crossword	10
End	Unit 6 Test	40
TOTAL		**180**

(!) SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

(★) TEACHING TIPS

- Start with the picture of textbook page 75: who has seen a video in a lesson? That is e-learning.
- Draw four big boxes on the board - e-learning, e-health, e-governance, e-commerce - and let pupils add the services of page 76.

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Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (35)	Ch 2 (57)	Ch 3 (48)	Test (40)	Total (180)
1						
2						
3						
4						
5						
6						
7						
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Practicals at a Glance

Workbook activities 6.1 - 6.7 of textbook pages 77 - 81

Practical (activity)	Students do	Check that ...
1 (6.1)	a family survey of the ICT services of the four fields	every service is in the right column (textbook page 76); pupils say how ICT saved time.
2 (6.2)	how a family with and a family without ready access to ICT get two services	pupils explain the digital divide in their own words. Keep the talk about societies, not homes.
3 (6.3)	find e-waste at home and at school; compare it with Figure 6.1	nobody opens, breaks or burns an item; broken CFL bulbs and leaking cells are left for an adult.
4 (6.4)	a poster: four toxic substances and the harm they do	each substance is linked to the right part of the body (Figure 6.2). Use the rubric below.
5 (6.5)	a 3R plan for the school computer lab	each item gets Reuse or Recycle; there is a rule for Reduce; the agency is a suitable one.
6 (6.6)	computer related job advertisements from newspapers	job titles and responsibilities are copied correctly and compared with the table of 6.7.
7 (6.7)	talk to a person who works in ICT; a short report	the report names the job, what the person does and the ICT subjects he or she studied.

SAFETY AND MATERIALS

- The textbook only gives the numbers of Workbook Activities 6.1 - 6.7; Practicals 1 - 7 follow them.
- Show only safe e-waste: an old keyboard, a mouse, a cable or a CD. Never open screens, CFL bulbs or cells.
- Newspapers with job advertisements (Practical 6); big sheets of paper for the posters (Practical 4).
- Ask parents beforehand if a pupil will interview a family member for Practical 7.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - ICT changes our society (6.1 - 6.3)	75 - 77	6.1, 6.2
Chapter 2 - E-waste and the 3R technique (6.4 - 6.6)	78 - 80	6.3, 6.4, 6.5
Chapter 3 - Computer related jobs (6.7)	80 - 81	6.6, 6.7
Unit 6 in One Page	82 (summary)	-

The textbook numbers its sections 6.1 (Impact of ICT on society, pages 75 - 80) and 6.2 (Computer related job opportunities, pages 80 - 81). The workbook numbers its own sections 6.1 - 6.7.

TIP: Ask which services the pupils' families use: paying a bill, online shopping, applying to a university ...

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What is e-learning?	**Teaching using ICT and the Internet**
2. Name one service of Hospital e-health.	**Diagnosing illnesses / storing patient information**
3. Name two services of e-governance.	**Bill payment, the Gazette (any two of page 76)**
4. What is e-commerce?	**Exchange of goods through the Internet; online shopping**
5. What is the digital divide?	**The gap between those who have ready access to ICT and those who have not**
6. Which waste is the most dangerous?	**Electronic waste (e-waste)**
7. Name three harmful e-waste items.	**e.g. CFL bulbs, electric cells, mobile phones**
8. Which toxic substance causes kidney weaknesses?	**Cadmium**
9. What are the 3Rs?	**Reduce, Reuse, Recycle**
10. Who installs and maintains a computer network?	**A Network Administrator**

> POSTER RUBRIC - PRACTICAL 4, 10 MARKS (OPTIONAL)

Mark the group posters of Workbook Activity 6.4 while each group explains its poster.

	The poster has ...	Marks
1	Four toxic substances, each named correctly (Figure 6.2).	2
2	Each substance is joined to the right part of the body.	4
3	The harm of each substance is written correctly.	2
4	One clear sentence: why we must never burn or dump e-waste.	1
5	The poster is neat, clear and easy to read.	1
	Total	**10**

ANSWERS

Chapter 1

ICT changes our society

A Which field?

8 marks

1. e-learning	**2.** e-governance	**3.** e-health	**4.** e-commerce
5. e-governance	**6.** e-learning	**7.** e-commerce	**8.** e-health

1 mark for each line with the tick in this column. Every sentence is one of the services of textbook page 76.

B Fill in the blanks

8 marks

1. e-learning	**2.** efficiently	**3.** anywhere	**4.** patient
5. e-governance	**6.** shopping	**7.** digital divide	**8.** behind

1 mark each. The extra words are **slowly** and **Gazette**.

C Write two services

8 marks

1. **School e-learning:** any two of - Clearer explanation with pictures and video clips; Collecting information through the Internet; Dialogue and messaging with other schools; The facility to learn anytime from anywhere.
2. **Hospital e-health:** any two of - Use of computer based modern equipment to diagnose illnesses; Facility to store patient information to be used when necessary.
3. **Public service - e-governance:** any two of - Bill payment; Specimen forms, loan information; Government Information Centre; Ability to apply to State universities through the Internet; To obtain circulars, legal services; Information in Gazette; The map of Sri Lanka.
4. **e-commerce:** any two of - Exchange of goods through the Internet; Online shopping and purchasing.

1 mark for each correct service (2 for each field). Accept the services in the pupil's own words.

D Choose the correct answer

5 marks

1. (b) e-learning	**2.** (c) storing patient information
3. (a) e-commerce	**4.** (d) online shopping
5. (a) ICT has not yet reached them	

E True or false

6 marks

1. **True** - ICT has brought about great changes everywhere, in every field.
2. **False** - e-learning means learning only from printed books.
3. **True** - Bill payment is a service of e-governance.
4. **False** - Hospitals cannot use computers to diagnose illnesses.
5. **False** - Every society in the world has ready access to computers and the Internet.
6. **True** - Online shopping and purchasing are part of e-commerce.

★ TEACHING TIP

- Let pupils sort cards of the ten e-waste items of Figure 6.1 on the board.

ANSWERS

Chapter 2

E-waste and the 3R technique

A Name the e-waste

8 marks

1. CFL bulbs	**2.** Screen with a Cathode Ray Tube	**3.** Motherboard	**4.** Mobile phones
5. Electric cells	**6.** Printer	**7.** CDs and covers	**8.** Keyboard and mouse

1 mark each. The extra names are **LED display screen** and **Cables**.

B Match the poison

6 marks

1. Selenium → C	**2.** Mercury → F
3. Arsenic → E	**4.** Chromium → A
5. Brominated Flame Retardants → D	**6.** Polychlorinated biphenyls → B

1 mark each (Figure 6.2).

C Which part of the body?

6 marks

1. Lead: **Brain**	**2.** Beryllium: **Lungs**	**3.** Trichloroethylene: **Liver**
4. Cadmium: **Kidneys**	**5.** Barium: **Brain**	**6.** Polyvinyl Chloride: **Lungs**

1 mark each (Figure 6.2). Burnt plastics harm breathing: the lungs.

D Reduce, reuse or recycle?

8 marks

1. Reduce	**2.** Reuse	**3.** Recycle	**4.** Reduce
5. Reuse	**6.** Recycle	**7.** Reduce	**8.** Reuse

1 mark each. Reuse = repair and use again; Recycle = hand it over.

E True or false

6 marks

1. **False** - E-waste makes the environment dirty, but it does not harm people.

2. **True** - Mobile phones and cables are harmful e-waste items.

3. **True** - Beryllium can cause lung cancer and breathing difficulties.

4. **False** - Burning plastics is a safe way to get rid of e-waste.

5. **False** - Reuse means handing e-waste over to suitable agencies.

6. **True** - The 3R concept can be used for the proper disposal of e-waste.

F Fill in the blanks

8 marks

1. dangerous	**2.** environment	**3.** Tube	**4.** CFL
5. Lead	**6.** Reduce	**7.** Reuse	**8.** agencies

1 mark each. The extra words are **Recycle** and **Mercury**.

G Choose the correct answer

5 marks

1. (c) electronic waste	**2.** (b) a CFL bulb
3. (a) skin diseases, weak nerves, cancer	**4.** (b) Reduce, Reuse, Recycle
5. (c) all living beings and the environment	

H Word search

10 marks

								S			
		F	U	R	A	N	S	N			
C		B	E	R	Y	L	L	I	U	M	M
H		L	E	A	D			X			E
R			S					O			R
O	A			E				I			C
M	R				L			D			U
I	S					E					R
U	E						N				Y
M	N	M	U	I	R	A	B	I			
	I								U		
	C	A	D	M	I	U	M			M	

Word	Starts at	Direction
LEAD	row 4, col 3	across
MERCURY	row 3, col 12	down
CADMIUM	row 12, col 2	across
ARSENIC	row 6, col 2	down
BARIUM	row 10, col 8	across, backwards
SELENIUM	row 5, col 4	slanting down
CHROMIUM	row 3, col 1	down
BERYLLIUM	row 3, col 3	across
DIOXINS	row 7, col 9	up
FURANS	row 2, col 3	across

1 mark for each word found and ticked.

ANSWERS

Chapter 3

Computer related jobs

A Match the jobs

8 marks

1. Software Quality Assurance Engineer → E	**2.** Software Architect → C
3. Network Administrator → F	**4.** Database Administrator → A
5. Web Application Developer → H	**6.** Graphic Designer → D
7. System Analyst → G	**8.** IT Consultant → B

B Who should do it?

6 marks

1. The computer network of the school does not work. **Network Administrator**
2. A shop needs banners and advertisements. **Graphic Designer**
3. A bank needs a person to look after its databases. **Database Administrator**
4. New software must be checked to work properly. **Software Quality Assurance Engineer**
5. A team needs a person to write computer programs. **Programmer**
6. A hospital wants advice on information technology. **IT Consultant**

1 mark each. For number 4 accept "Quality Assurance Engineer".

C True or false

6 marks

1. **True** - People with a proper education on ICT related subjects have ample job opportunities.
2. **False** - A Graphic Designer installs and maintains computer networks.
3. **True** - A Programmer develops computer programs using programming languages.
4. **False** - A Database Administrator designs banners and magazines.
5. **True** - An IT Consultant advises organizations on matters related to information technology.
6. **False** - A Software Architect only repairs keyboards and mice.

D Choose the correct answer

5 marks

1. (b) ample job opportunities	**2.** (a) a Software Engineer
3. (b) a System Analyst	**4.** (b) websites
5. (a) a Software Architect	

E My dream ICT job

4 marks

1. **Which computer related job would you like to do? (1 mark)**
   Any one of the ten jobs of 6.7 (1 mark).
2. **What does a person in that job do? (2 marks)**
   The responsibilities of that job, as in the table of 6.7 (2 marks; 1 mark if only half of it).
3. **Why must you study ICT related subjects well for this job? (1 mark)**
   Those who have a proper education on ICT related subjects have ample job opportunities (1 mark).

TIP: Link Figure 6.2 to Science: the brain, the lungs, the liver and the kidneys.

F Answer these

4 marks

1. What does a System Analyst do? Write two things. (2 marks)

A System Analyst analyses organizational needs and designs computer systems (1 mark), and coordinates the development of software (1 mark).

2. What is the difference between a Programmer and a Software Architect? (2 marks)

A Programmer develops computer programs using programming languages (1 mark). A Software Architect designs the architecture of software - its components and their interconnections (1 mark).

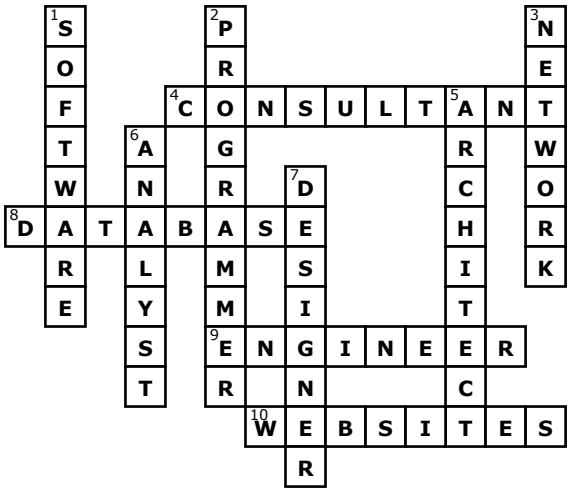
G Read the advertisement

5 marks

1. Which job does the advertisement offer? **Network Administrator**
2. What will this person do at the bank? **Install, configure and maintain the computer network of the bank (2 marks; 1 mark for two of the three verbs)**
3. What education must this person have? **A proper education on ICT related subjects**
4. The bank also needs a person to plan, install and maintain its databases. Name that job. **Database Administrator**

H Crossword

10 marks



Across: 4 CONSULTANT, 8 DATABASE, 9 ENGINEER, 10 WEBSITES

Down: 1 SOFTWARE, 2 PROGRAMMER, 3 NETWORK, 5 ARCHITECT, 6 ANALYST, 7 DESIGNER

TEACHING TIPS

- Talk about the digital divide between societies and countries, never about one pupil's home.
- Show the digital divide picture and ask: what can the people on the right do that the people on the left cannot?
- Bring a few safe e-waste items to class: an old keyboard, a mouse, a cable, a CD. Never broken screens, CFL bulbs or leaking cells.

MARKING SCHEME

Unit 6 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) efficiently	**2.** (a) information
3. (a) developing	**4.** (d) a ruler
5. (a) e-waste	**6.** (a) software
7. (a) jobs	**8.** (b) A and C

2 Fill in the blanks

8 marks (1 each)

1. behind	**2.** equipment	**3.** electronic	**4.** mobile
5. Programmer	**6.** Database	**7.** Analyst	**8.** Web

3 Name the pictures

8 marks (1 each)

1. Public service - e-governance	**2.** School e-learning	**3.** e-commerce	**4.** Hospital e-health
5. Screen with a Cathode Ray Tube	**6.** CFL bulbs	**7.** CDs and covers	**8.** Motherboard

Cards 1 - 4: accept e-governance, e-learning, e-commerce and e-health. Cards 5 - 8 are items of Figure 6.1.

4 Answer these

16 marks

1. What is e-learning? Write one way it helps students. (2 marks)

Teaching using ICT and the Internet (1 mark). Any one: a clearer explanation with pictures and video clips; collecting information through the Internet; messaging with other schools; learning anytime from anywhere (1 mark).

2. Write two services of Public service - e-governance. (2 marks)

Any two (1 mark each): bill payment; specimen forms, loan information; the Government Information Centre; applying to State universities through the Internet; circulars, legal services; information in Gazette; the map of Sri Lanka.

3. What is the digital divide? (2 marks)

Some societies have ready access to computers and the Internet while others do not (1 mark); the gap between those who have and those who have not is the digital divide (1 mark).

4. Why is electronic waste the most dangerous waste? (2 marks)

It makes the environment dirty (1 mark) and it harms people with toxic poisons (1 mark).

5. Name two toxic substances in e-waste and the harm that each one does. (2 marks)

Any two from Figure 6.2 with the right harm (1 mark each), e.g. Lead - defects in the brain, stomach and blood circulation; Mercury - nerves, liver; Cadmium - kidney weaknesses; Beryllium - lung cancer.

6. Explain the 3R technique. (3 marks)

Reduce: minimize the purchase or collection of unnecessary goods (1 mark). Reuse: use again and again after repairing, wherever possible (1 mark). Recycle: hand over to suitable agencies for recycling (1 mark).

7. Write what each of these people does: a Network Administrator, a Graphic Designer and an IT Consultant. (3 marks)

Network Administrator: installs, configures and maintains computer networks (1 mark). Graphic Designer: develops web pages, advertisements, magazines and banners using software (1 mark). IT Consultant: advises organizations on matters related to information technology (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They put bill payment under e-commerce. In the textbook it is a public service: **e-governance**.
- They think the digital divide is a quarrel between countries. It is the gap in **ready access** to computers and the Internet.
- They mix up Reuse (repair and use again) and Recycle (hand over to a suitable agency).
- They think e-waste only makes the environment dirty. It also **harms people** with toxic poisons.
- They mix up Arsenic and Dioxins and Furans: both cause cancer, nerve problems and skin diseases.
- They mix up the Web Application Developer (websites) and the Graphic Designer (web pages, advertisements, magazines, banners).
- They misspell the long names: Beryllium, Trichloroethylene, Polychlorinated biphenyls.

HOMEWORK AND EXTRA PRACTICE

- List the ICT services your family used this month under the four fields of 6.2.
- Find two old electronic items at home. Which R of the 3R technique fits each one?
- Cut out an ICT job advertisement. Write the job title and its responsibilities.
- Fast learners: prepare a two-minute talk, "How ICT has changed our village".